

ASSIGNMENT 1: Mathematics for Engineers I: MAT 8166: Year 1 Computer Engineering

1. Determine the range of the following function

$$f(x) = x^2 - 4x + 11.$$

2. Determine the domain of definition of the functions

$$\begin{aligned} i. f(x) &= \frac{x^2 + x - 12}{\sqrt{3 - 5x}} \\ ii. f(x) &= \frac{\sqrt{x^2 - 9x + 18}}{x^2 + 4x - 21}. \end{aligned}$$

3. Calculate the following limits

$$\begin{aligned} i. A &= \lim_{x \rightarrow 0} \frac{2x + \sin 2x}{x + \sin 5x} \\ ii. B &= \lim_{x \rightarrow 0} \left(\frac{1}{\sin^3 x} - \frac{1}{x^3} \right) \end{aligned}$$

4. A company that produces cell phones has a cost function of $C = x^2 - 1200x + 36400$, where C is cost in dollars and x is number of cell phones produced (in thousands). How many units of cell phone (in thousands) minimizes this cost function?
5. A country launches a clean energy project aimed at reducing carbon emissions by replacing diesel buses with electric buses. In 2020, the country's public transport emitted 50,000 tons of CO₂. Every year since then, emission levels have been reduced by 4,000 tons due to the new policies.

Let $E(t)$ represents the total CO₂ emissions (in tons) after t years since 2020.

- (a) Formulate a linear function that models $E(t)$.
 - (b) Use the model to predict the emissions in 2025.
 - (c) In what year will the country reach zero emissions if the current rate continues?
6. A processor's performance is modeled by

$$f(v) = k\sqrt{v - 0.8},$$

where v is voltage and $f(v)$ the clock frequency.

Tasks:

- (a) Find the domain of $f(v)$.
 - (b) If $0.8 \leq v \leq 1.5$, determine the range.
7. A chip temperature is modeled as

$$T(p) = 70 + 15 \ln(p),$$

where p is power in watts.

Tasks:

- (a) Determine the domain.
- (b) Find the range for $1 \leq p \leq 20$.

8. A digital filter outputs

$$A(t) = \frac{5}{1 + e^{-2t}}.$$

Tasks:

- (a) Determine the domain.
- (b) Determine the range.
- (c) Interpret the range physically.

9. The temperature of a processor after shutdown obeys Newton's cooling law:

$$T(t) = 25 + (90 - 25)e^{-0.08t}.$$

Tasks:

- (a) Compute $T(10)$.
- (b) Find t such that $T(t) < 40$.

10. The running time of a search algorithm is

$$T(n) = 5 \log_2(n) + 12.$$

Tasks:

- (a) Compute $T(1024)$.
- (b) Solve $T(n) = 32$.
- (c) Explain slow growth mathematically.